



Report

Nara Avila Moraes

Physics Institute
IFUSP

2026 / 07 / 03

Summary

① Tasks Status

② Results

③ Selected Article

General Tasks Status

Task	Status	
2026-06 RETINHA presentation	✓	Link
2026-06 Scientific report	🕒	Link

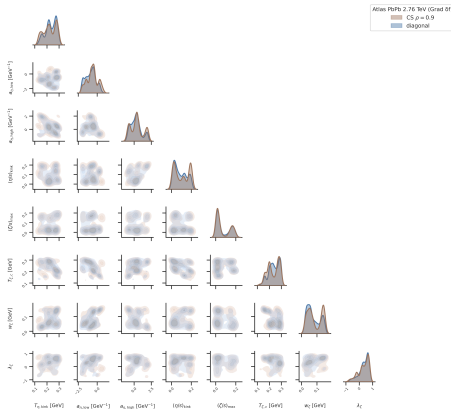
Task	Status	
Journal Club - Automations	🕒	

Correlated Uncertainty Study Tasks Status

Task	Status	Link
Alice ins1222333 Studies		
↪ Mcmc Params	✓	Plots
↪ RDF Model	✓	Plots
↪ CS Model	✓	Plots
↪ None Uncertainty Correlation	🔄	Could not initialize ptemcee walkers inside posterior support after 1000 attempts. Try a different seed or relax the initial ranges.
Atlas ins1759506 studies		
↪ Uncertainty correlation	🔄	
RHIC Au-Au/d-Au Studies JETSCAPE		

Baseline vs CS correlation model

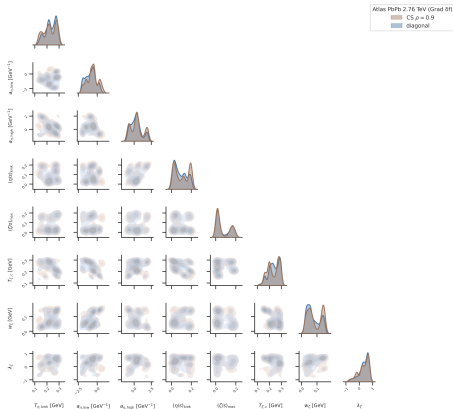
Bayesian uncertainty-correlation study: CS vs diagonal



Figure

Baseline vs RDS correlation model

Bayesian uncertainty-correlation study: CS vs diagonal



Figure

Selected Article

Hadronic tensor in lattice gauge theories by quantum computing

Dairui Zou,^{1,2,3} Tianyin Li,^{4,*} Jian Liang,^{1,3,†} Enke Wang,^{1,3,‡} and Hongxi Xing^{1,3,5,§}
(QuNu Collaboration)

¹*State Key Laboratory of Nuclear Physics and Technology, Institute of Quantum Matter,
South China Normal University, Guangzhou 510006, China*

²*Key Laboratory of Atomic and Subatomic Structure and Quantum Control (MOE),
Guangdong-Hong Kong Joint Laboratory of Quantum Matter, Guangzhou 510006, China*

³*Guangdong Basic Research Center of Excellence for Structure and Fundamental Interactions of Matter,
Guangdong Provincial Key Laboratory of Nuclear Science, Guangzhou 510006, China*

⁴*RIKEN Center for Interdisciplinary Theoretical and Mathematical Sciences (iTHEMS), RIKEN, Wako 351-0198, Japan*

⁵*Southern Center for Nuclear-Science Theory (SCNT),
Institute of Modern Physics, Chinese Academy of Sciences, Huizhou 516000, China*

(Dated: June 16, 2026)

The hadronic tensor encodes crucial information regarding the internal structure of hadrons, reflecting the non-perturbative features of quantum chromodynamics (QCD). In this work, we directly compute the hadronic tensor within (1+1)-dimensional U(1) and SU(2) gauge theories by evaluating real-time current-current correlation functions. Utilizing quantum algorithms executed on classical hardware, we demonstrate that the hadron form factors for both meson and baryon states can be reliably extracted from the hadronic tensor. Our methodology is validated by strong agreement with both direct calculation and exact diagonalization of the form factors.

I. INTRODUCTION

Figure: Lattice Quantum Chromo Dynamics (QCD) calculations on Quantum Computers

15 Jun 2026